MIL-E-1/49E
27 April 1972
SUPERSEDING
MIL-E-1/49D
12 February 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AS7G

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: High current, low Mμ, double triode

Outline --- 4-9

Base --- B8-11

Envelope --- ST16

Cathode --- Coated unipotential

Base connections:

Pin no.	1	2	3	4	5	6	7	8
Element ---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Ib/p	Pp/p	Ehk	Rk/k	TE	Alt
Unit:	V	Vdc	Vdc	mAdc	W	v	Ohms	°C	ft
Maximum:	6.9	275	---	125	14.0	330	--	200	10,000
Minimum:	5.7	---	---	---	----	---	--	---	----
TEST CONDITIONS:	6.3	135	0	---	----	---	--	---	----

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Qualification							
1216	Base material insulating quality		----	---	---	----	----	---
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)	See note 3	0.65	II	Ib	100	150	mAdc
1256	Electrode current (2) (anode)	Ec = -200 Vdc; Eb = 250 Vdc See note 3	0.65	II	Ib	----	10	mAdc
1266	Total grid current	Rg = 1.0 meg See notes 2 and 4	0.65	II	Ic	0	-4.0	μAdc
1231	Emission current	Eb = Ec = 10 Vdc See note 5	0.65	II	Is	75	----	mAdc
1201	Shorts and discontinuity detection		0.4	II	---	----	----	---
	Quality conformance inspection, part 2							
1301	Heater current		6.5	S3	If	2.26	2.74	A
1336	Heater-cathode leakage	See note 1	6.5	S3	Ihk	----	50	μAdc
1031	Low-frequency vibration	Rp = 2,000 ohms; Ec = -7 Vdc; Rk = 0 See note 1	6.5	S3	Ep	----	200	mVac
1306	Transconductance	See note 3	6.5	S3	Sm	5800	8200	μmhos
1316	Amplification factor	See note 3	6.5	S3	M	1.4	2.6	---
1105	Permanence of marking		----	---	---	----	----	---
1101	Secureness of base, base insert, and cap		6.5	S3	---	----	----	---
1111	Base pin solder depth		6.5	S3	---	----	----	---
	Quality conformance inspection, part 3							
1501	Intermittent life-test (500 hours)	Group A; Rk = 125; Rg = 1 meg; Ehk = -300 Vdc See note 1	----	---	---	----	----	---
----	Intermittent life-test end points (500 hours)							
1306	Transconductance		----	---	Sm	4900	----	μmhos
1266	Total grid current		----	---	Ic	----	-10	μAdc
1336	Heater-cathode		----	---	Ihk	----	100	μAdc

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NOTES:

1. Tie 1a to 2a; 1g to 2g; 1k to 2k
2. With both units operating, Ic is the sum of I1c and I2c.
3. With both units operating, read each unit separately.
4. This test to be performed at the conclusion of the holding period.
5. Read each unit separately: ground elements of opposite side; Rk = 0.

Custodians:

Army - EL
Navy - EC
Air Force - 80

Review activities:

Army - EL
Navy - EC
Air Force - 11, 17

User activities:

Army - MU
Navy - WP, MC, CG
Air Force - None

Preparing activity:

Air Force - 80

(Project 5960-2524)

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NOTICE OF VALIDATION

INCH-POUND

MIL-E-1/49E
NOTICE 1
14 July 1989

MILITARY SPECIFICATION

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TYPE 6AS7G

Military Specification MIL-E-1/49E dated 13 APRIL 1984 has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Preparing Activity:
Air Force - 85

Review activities:
Army - ER
Navy - EC
Air Force - 17,99

User activities:
Army - MU
Navy - MC, CG
Air Force - 11

AMSC N/A

FSC 5960

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